

IntelliMeet Pro: AI Meeting Assistant & Workflow Management

Software Re-Engineering

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1. *Introduction*

Software systems evolve over time, and maintaining their efficiency, scalability, and usability requires systematic improvement. Software re-engineering is a disciplined process that involves analyzing an existing system, understanding its structure and behavior, and improving it without changing its core functionality.

This project focuses on the re-engineering of **IntelliMeet Pro**, an AI-powered meeting automation platform. The system is analyzed through reverse engineering techniques to extract architectural and design knowledge. Based on this understanding, weaknesses are identified and improvements are proposed.

The objective of this report is to:

- Understand the existing system
- Recover system design
- Identify limitations
- Propose re-engineering improvements

2. *Selected Application*

IntelliMeet Pro is a cloud-based AI meeting assistant designed to automate meeting workflows and convert discussions into structured knowledge.

2.1. *Key features*

- Automatic meeting bot joining (Zoom, Google Meet, Teams)
- Audio recording and storage
- Speech-to-text transcription using AI
- Speaker diarization (identifying speakers)
- AI-based meeting summarization
- Action item extraction and assignment
- Semantic search using vector databases
- Slack bot integration for queries
- Integration with Jira, Trello, Asana
- Subscription and billing system

2.2. *Target users*

- Organizations and corporate teams
- Project managers and team leads
- Developers and knowledge workers
- Remote and hybrid teams

3. *System Exploration*

3.1. *System overview*

IntelliMeet Pro transforms traditional meetings into structured, searchable, and actionable insights. The system automatically joins meetings, records audio, processes it using AI, and provides summaries and action items.

It eliminates manual note-taking and improves productivity by bridging the gap between discussions and execution.

3.2. *Main functionalities*

1. User Management

- Registration and login
- Email verification
- Role-based access control (Admin, Member)

2. Workspace Management

- Create workspaces
- Invite members
- Assign roles

3. Meeting Management

- Schedule meetings
- Deploy AI bot
- Record meeting audio

4. Transcription

- Convert speech to text
- Timestamped transcript generation
- Speaker identification

5. Summarization

- AI-generated summaries
- Key points and decisions extraction

6. Action Item Extraction

- Identify tasks from discussions
- Assign tasks to users

7. Semantic Search

- Natural language queries
- Retrieve relevant meeting content

8. Integrations

- Jira, Trello, Asana
- Slack bot interface

9. Subscription System

- Multiple pricing tiers
- Payment processing

4. Reverse Engineering Analysis

4.1. System Architecture

The system follows a Layered Architecture deployed on cloud infrastructure.

1. Presentation Layer

- Web application
- Slack bot interface

2. Application Layer

- .NET backend API
- Business logic handling

3. AI/NLP Layer

- Whisper (transcription)
- GPT (summarization)
- Translation services

4. Data Layer

- SQL database (structured data)
- Azure Blob Storage (audio files)
- Pinecone (vector embeddings)

5. Integration Layer

- External APIs (Zoom, Slack, Stripe, etc.)

4.2. System Modules

The system is divided into the following modules:

1. Authentication Module
2. Workspace Management Module
3. Meeting Management Module
4. Transcription Module
5. Summarization Module
6. Action Item Module
7. Search Module (RAG-based)
8. Integration Module
9. Subscription Module
10. Notification Module

4.3. Database Entities

Key entities include:

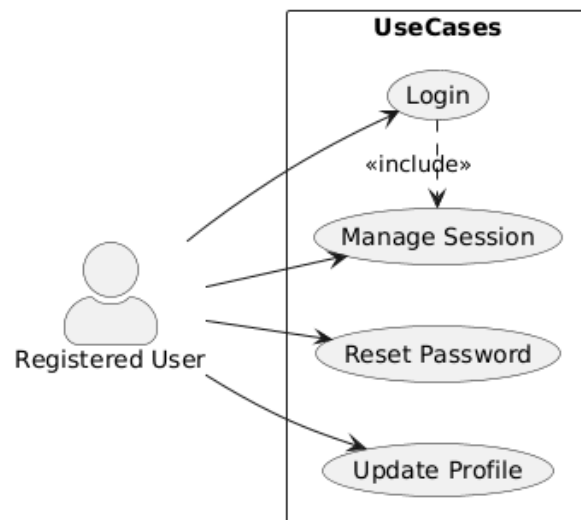
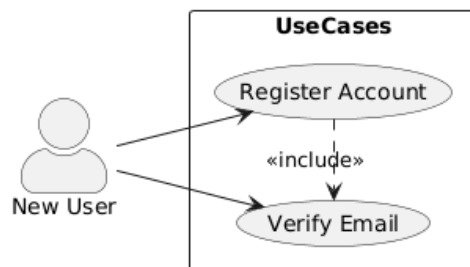
1. **Users** → stores user information
2. **Workspaces** → groups users and meetings
3. **Meetings** → meeting metadata
4. **Transcripts** → speech-to-text data
5. **Summaries** → AI-generated summaries
6. **ActionItems** → extracted tasks
7. **SpeakerProfiles** → voice identity tracking
8. **Integrations** → API tokens
9. **Subscriptions** → billing details

5. UML Diagrams

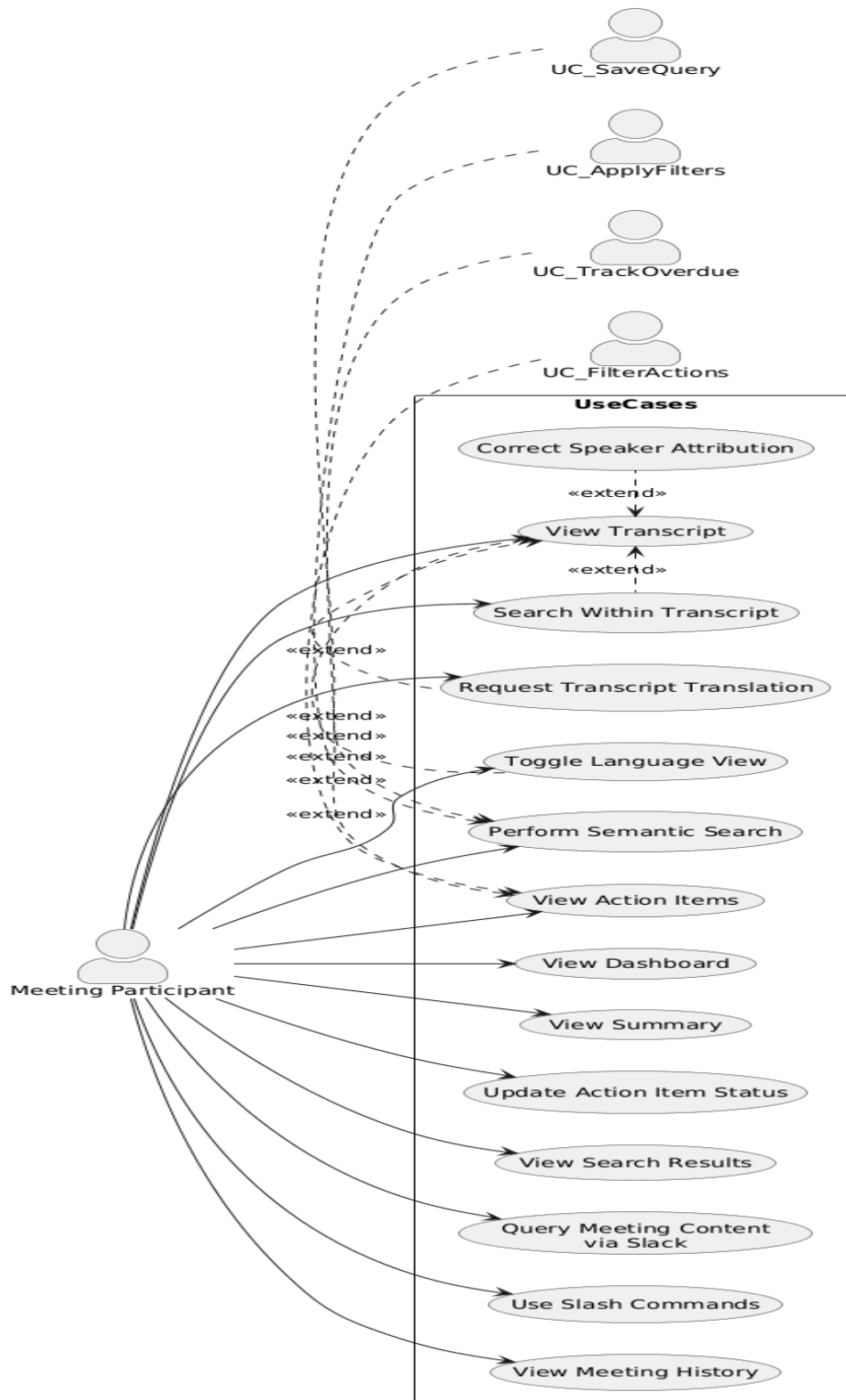
5.1. Use Case Diagram

Actors and Use Cases:

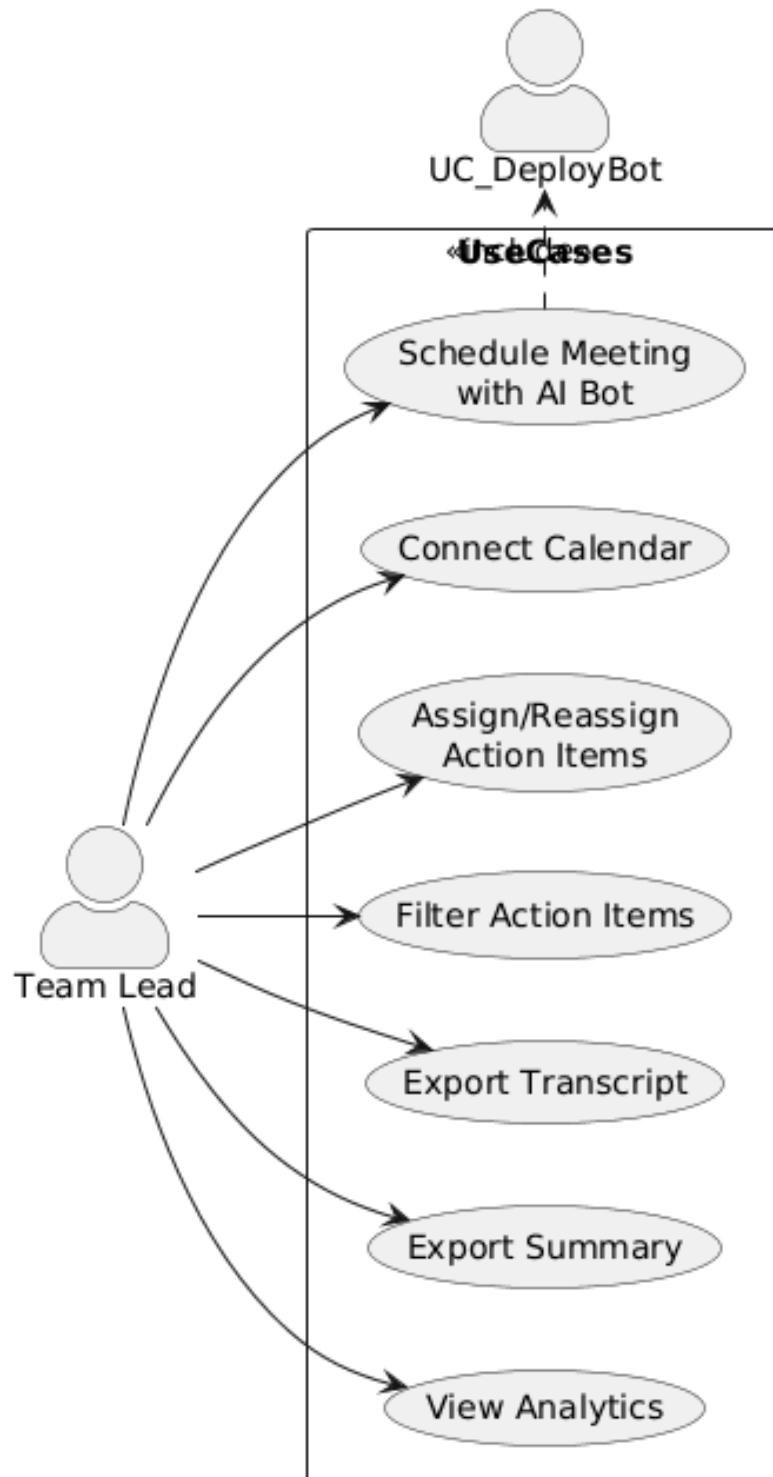
- New User / Registered User: Register/Login



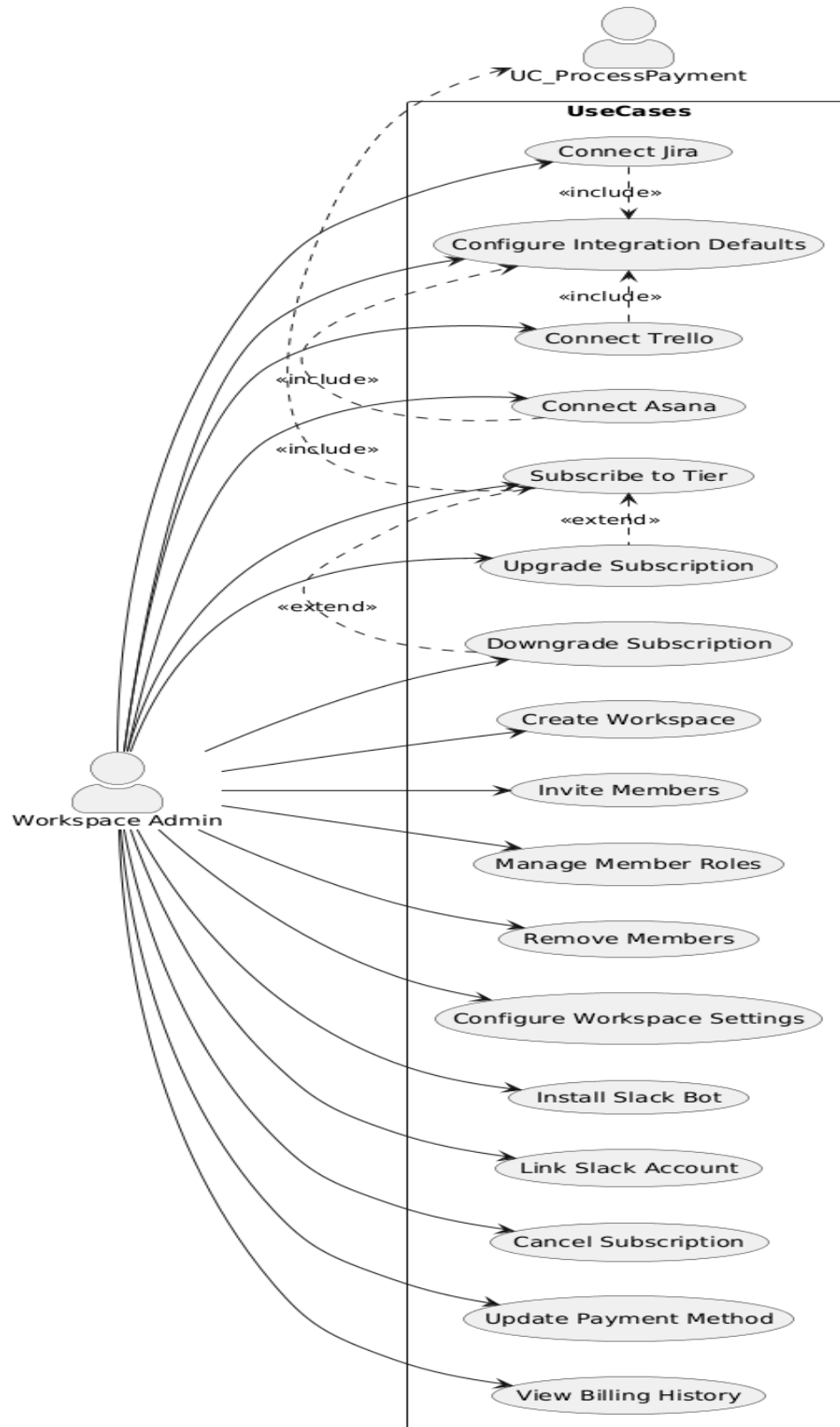
- Meeting Participant: Join Meeting via Bot



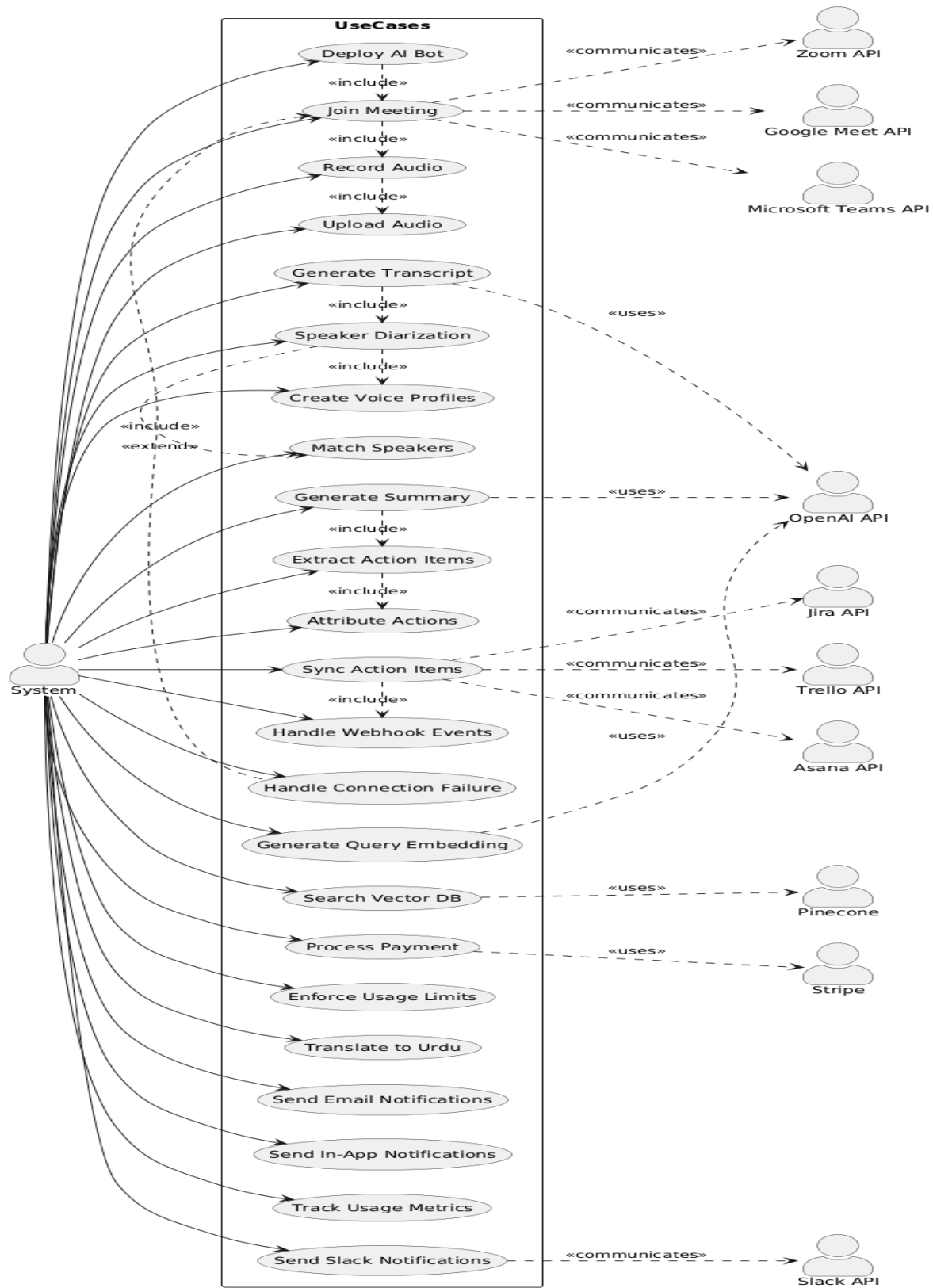
- Team Lead:Schedule Meeting



- Workspace Admin:Create Workspace



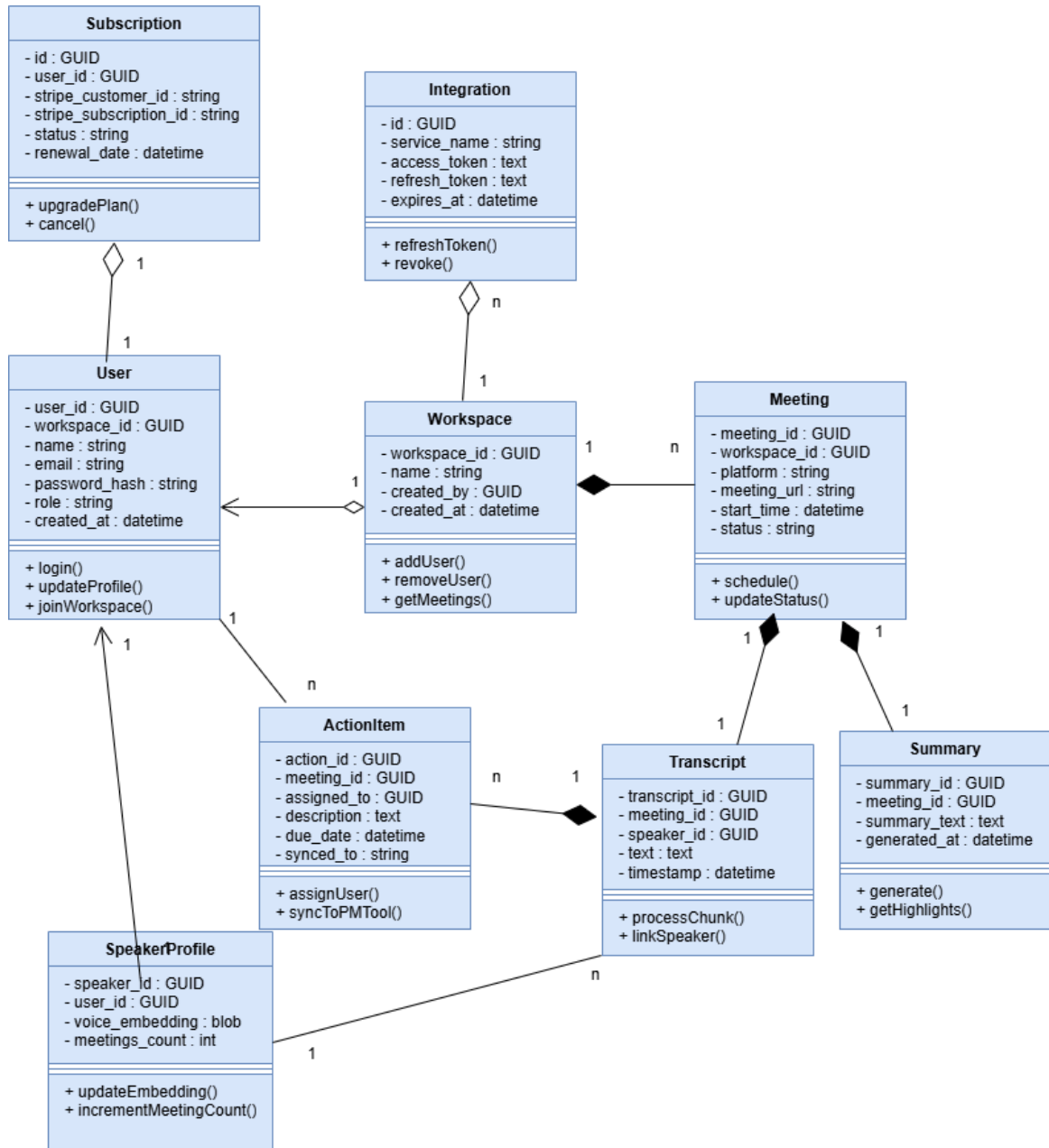
- System:Generate Transcript, Generate Summary, Extract Action Items



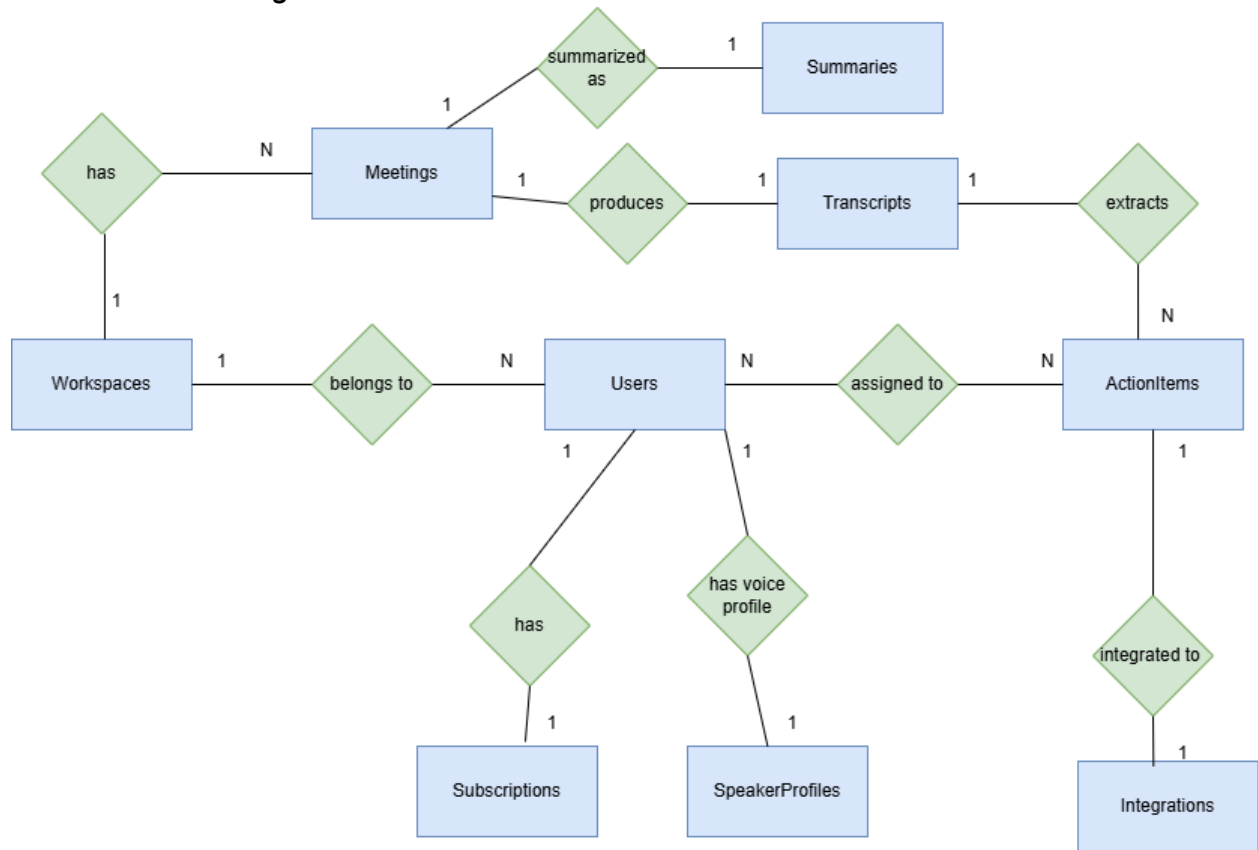
5.2. Domain Model

5.2.1. Class Diagram

- User → belongs to Workspace
- Workspace → contains Meetings
- Meeting → has Transcript
- Meeting → has Summary
- Summary → generates ActionItems
- Meeting → linked to SpeakerProfiles



5.2.2. *ER Diagram*



5.3. Data Flow Diagram

5.3.1. Level 0 Context Diagram

External Entities:

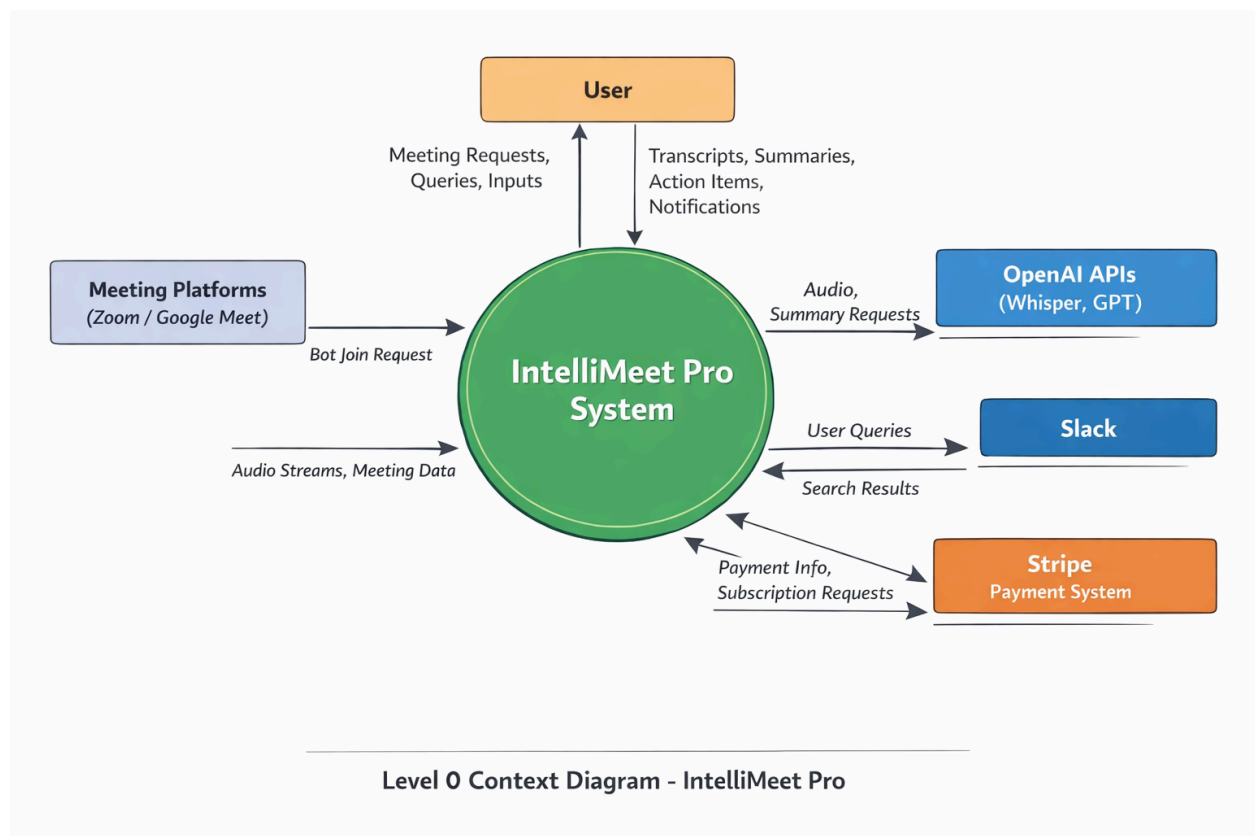
- User
- Meeting Platforms
- OpenAI APIs
- Slack
- Stripe

System:

- IntelliMeet Pro

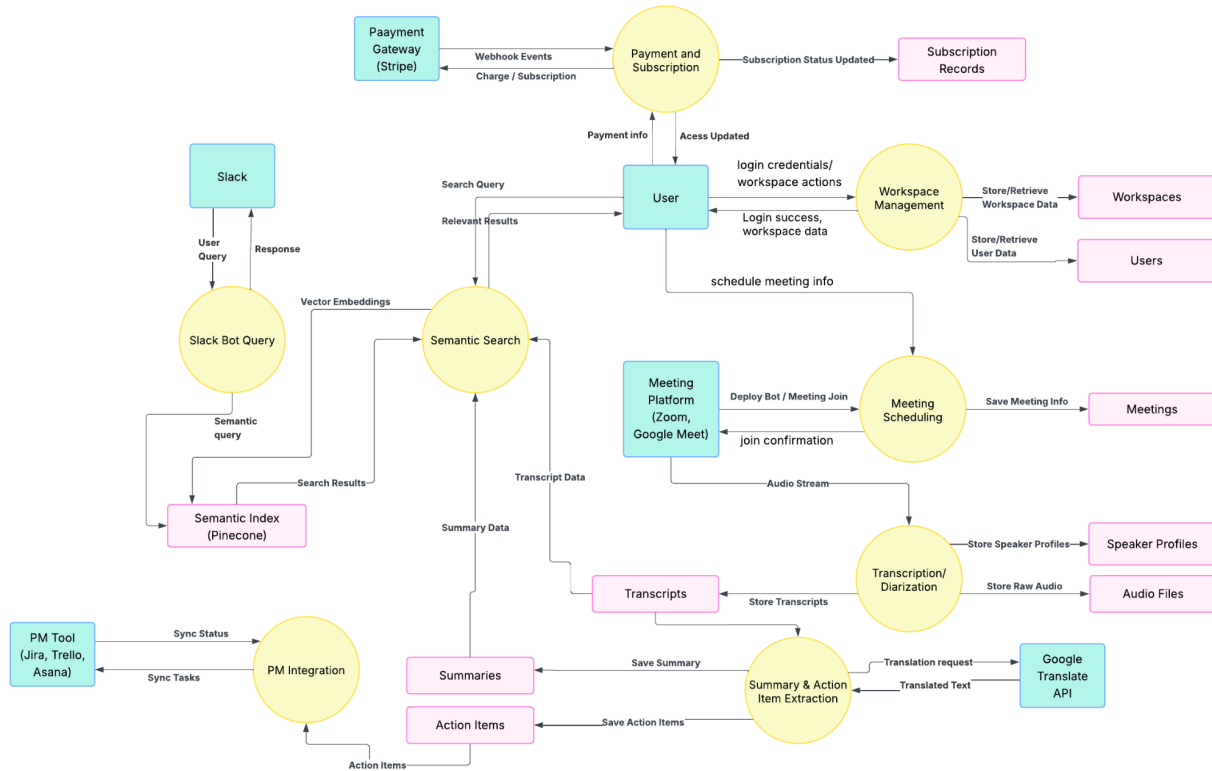
Data flows:

- Meeting data input
- Processed outputs (transcripts, summaries, tasks)



5.3.2. Level 1 DFD

This Level-1 DFD shows how IntelliMeet Pro processes meeting data end-to-end: the user authenticates, manages their workspace, and schedules meetings, which triggers the bot to join external meeting platforms. Audio from the meeting is sent to the transcription/diarization module, producing transcripts, speaker profiles, and audio files that feed into the summary and action item extraction process. Summaries and action items are stored and used by the semantic search engine, which also powers Slack bot queries. Action items are synced with external project management tools, while the payment system manages subscriptions through Stripe and updates access permissions.



6. Weakness Identification

1. Dependency on External APIs

The system heavily relies on APIs such as OpenAI, Zoom, and Slack. Any API changes or downtime can disrupt functionality.

2. Lack of Real-Time Transcription

The system processes transcription after meetings, limiting real-time usability.

3. Complex Multi-Storage Architecture

Using SQL, Blob Storage, and Pinecone increases system complexity and synchronization challenges.

4. High Operational Cost

AI services, storage, and integrations make the system expensive to maintain.

5. AI Accuracy Issues

Summaries and action items depend on AI, which may produce incorrect or misleading results.

6. Performance Bottlenecks

The RAG pipeline may slow down as data grows.

7. Security Concerns

Sensitive meeting data is stored without end-to-end encryption.

8. No Mobile Application

The system lacks native mobile support, reducing accessibility.

7. ***Re-Engineering Proposal***

1. Architecture Improvement

Transition from monolithic architecture to microservices architecture:

- Independent services for AI, meetings, search, etc.
- Improves scalability and maintainability

2. Real-Time Transcription

Implement streaming transcription:

- Live captions during meetings
- Enhanced user experience

3. AI Reliability Enhancement

- Add confidence scores
- Allow user corrections
- Validate outputs

4. Data Architecture Optimization

- Introduce unified data access layer
- Improve synchronization between databases

5. Security Enhancement

- End-to-end encryption
- Secure storage policies
- Audit logging

6. Performance Optimization

- Cache frequently accessed data
- Optimize embedding generation
- Use indexing strategies

7. API Resilience

- Implement fallback mechanisms

- Use adapter pattern for integrations
8. Mobile Application Development
 - Develop mobile app using React Native or Flutter
 9. Cost Optimization
 - Optimize API usage
 - Introduce efficient batching and caching

8. Conclusion

This project successfully applied software re-engineering principles to IntelliMeet Pro by analyzing its architecture, recovering its design, identifying weaknesses, and proposing improvements.

The system demonstrates strong capabilities in AI-driven meeting automation but suffers from complexity, dependency risks, and scalability challenges. The proposed re-engineering solutions aim to enhance performance, reliability, and user experience.

Through this exercise, the importance of reverse engineering and system improvement in real-world software systems is clearly demonstrated.